

Constructing an Immigration Shift-Share Instrument

Project Exercises — Path 2

Jessica Ding

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1 Introduction

This Advanced Econometrics project constructs a shift-share (Bartik) instrumental variable for immigration across U.S. commuting zones using five decades of Census microdata (1970–2010), following the approach of Kim et al(2024). The instrument is then used to estimate the causal effect of immigration on local wages via two-stage least squares (2SLS). The full pipeline of the project from raw IPUMS microdata to regression output is implemented in Stata do-file and is fully reproducible given the raw data. Revised do-files are also available,so in total there are 4 do files(main,fix 2010, fix step 2 to 5, fix step6) used to finish this.

2 Data

2.1 Sources

Individual-level microdata come from the U.S. Decennial Census and American Community Survey, accessed through IPUMS USA (5) for five decades: 1970, 1980, 1990, 2000, and 2010. All five samples were downloaded via the IPUMS API using `ipumspy`. Geography is harmonized to 1990-definition commuting zones using crosswalks from Dorn (2009). Birthplace codes are mapped to approximately 82 origin-country groups using the crosswalk provided by Kim et al. (2024).

2.2 Sample Restrictions

For each decade I impose the following restrictions: (1) drop Alaska and Hawaii (`statefip` $\in \{2, 15\}$); (2) drop individuals in non-standard group quarters (`GQ` $\in \{0, 3, 4\}$); (3) drop individuals under age 16. Geography is converted to commuting zones using a many-to-many join (`joinby`), with person weights adjusted by the allocation factor supplied in each Dorn crosswalk ($\tilde{w}_i = \text{afact} \times \text{perwt}$).

2.3 Wage Construction

Hourly wages are computed as annual wage income divided by annual hours worked. Annual hours are the product of weeks worked and usual hours per week. For 1970 and 1980, weeks worked (`wkswork2`) is recorded in intervals; I convert to continuous values using midpoint imputation (e.g., the interval “40–47 weeks” is assigned 43.5 weeks). Wage observations at the top 1% of the weighted distribution are dropped to limit the influence of top-coded incomes.

3 Empirical Strategy

3.1 Shift-Share Instrument

Local immigration seem to be endogenous to local labor demand: immigrants may sort toward booming areas, biasing OLS estimates of the wage effect upward (or downward in a labor demand framework). Following Card (2001) and Kim et al. (2024), I construct a shift-share (Bartik) instrument that predicts local immigration from *national* inflows by origin country, distributed across commuting zones using *historical* (1970) settlement patterns.

The instrument for commuting zone c in decade t is:

$$\tilde{B}_{c,t} = \frac{1}{\text{pop}_{c,1970}} \sum_o \underbrace{\frac{\text{imm}_{c,o,1970}}{\text{imm}_{o,1970}^{\text{nat}}}}_{\text{1970 share}} \times \underbrace{(\text{imm}_{o,t}^{\text{nat}} - \text{imm}_{o,t-1}^{\text{nat}})}_{\text{national shift}} \quad (1)$$

where $\text{imm}_{c,o,1970}$ is the number of immigrants from origin o living in commuting zone c in 1970, $\text{imm}_{o,1970}^{\text{nat}}$ is the total U.S. immigrant population from origin o in 1970, and $\text{pop}_{c,1970}$ is the total 1970 population of commuting zone c .

3.2 Identification

The instrument is valid if the 1970 settlement shares are uncorrelated with decade- t local labor-demand shocks, conditional on controls. This is plausible because immigrants' location choices in 1970 reflect historical network formation rather than current economic conditions.

3.3 Estimating Equations

First stage:

$$\Delta\text{imm}_{c,t} = \beta \tilde{B}_{c,t} + \gamma_t + \varepsilon_{c,t} \quad (2)$$

where $\Delta\text{imm}_{c,t} = (\text{imm}_{c,t} - \text{imm}_{c,t-1})/\text{pop}_{c,1970}$ is the actual immigration change normalized by 1970 population, and γ_t are decade fixed effects. Standard errors are clustered at the commuting-zone level.

Wage regression:

$$\Delta\text{wage}_{c,t} = \alpha \Delta\text{imm}_{c,t} + \gamma_t + u_{c,t} \quad (3)$$

I estimate this equation by OLS and by IV/2SLS, instrumenting $\Delta\text{imm}_{c,t}$ with $\tilde{B}_{c,t}$. Both specifications include year fixed effects and CZ-clustered standard errors. I also report specifications that add commuting-zone fixed effects.

4 Results

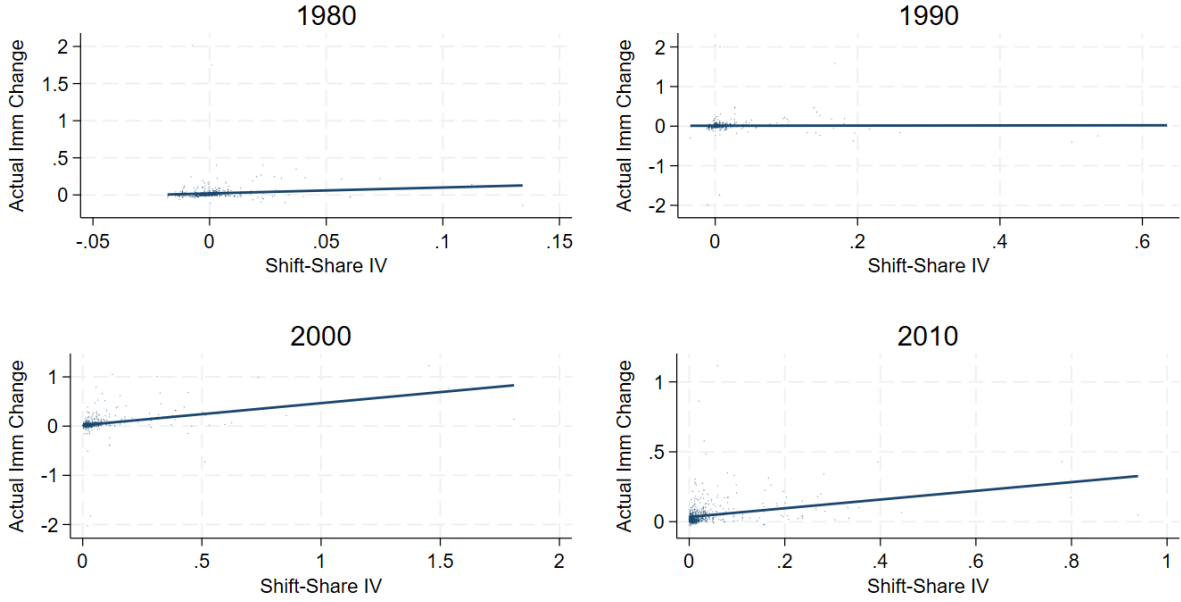
4.1 Summary Statistics

Table reports means and standard deviations of the key variables by decade. The shift-share instrument and actual immigration change track each other closely, consistent with a strong first stage. Both grow over the sample period, reflecting rising immigration to the United States from 1970 to 2000, with a modest decline in the 2010 decade.

4.2 First Stage

Figure 1 below plots the shift-share instrument against the actual immigration change for each decade. The positive relationship is visible in all four panels, with a more steep slope in later decades when immigration was more concentrated in historically immigrant-receiving commuting zones.

First Stage: Shift-Share IV vs. Actual Immigration Change



Each panel shows one decade. Fitted line from OLS.

Figure 1: First-stage scatter plots by decade. Each point is a commuting zone. The fitted line is from OLS.

Table 1 below reports the first-stage regression. The instrument enters with a large, statistically significant coefficient in all specifications. As is shown, the F-statistic exceeds the conventional threshold of 10, indicating that the instrument is strong.

Table 1: First Stage: IV to Actual Immigration Change

	(1)	(2)
	(1) Year FE	(2) Year+CZ FE
imm_shift	0.384***	0.457***
	(0.097)	(0.121)
<i>N</i>	2775	2772
CZ FE	No	Yes
Year FE	Yes	Yes

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

4.3 Wage Regressions

Table 2 reports OLS and IV estimates of the effect of immigration on local wages. The OLS estimate (column 1) is 0.743 (s.e. = 0.446), suggesting that a one-unit increase in the

immigration rate is associated with a \$0.74 increase in mean hourly wages. The IV estimate (column 2) is 0.775 (s.e. = 0.896), very close to OLS. The first-stage F-statistic is 16.15, well above the conventional threshold of 10, confirming instrument strength.

When commuting-zone fixed effects are added (columns 3 and 4), the estimates rise substantially. The IV+FE estimate is 4.073 (s.e. = 2.392), significant at the 10% level, with a first-stage F of 8.23.

The similarity between OLS and IV in the baseline specification suggests limited endogeneity bias once year fixed effects absorb aggregate demand variation. The larger IV estimate with CZ fixed effects is consistent with immigrants sorting into slower-growing commuting zones within the panel: controlling for CZ-level trends reveals a larger positive wage effect of immigration.

Table 2: Effect of Immigration on Wages

	(1)	(2)	(3)	(4)
	(1) OLS	(2) IV	(3) OLS+FE	(4) IV+FE
imm_actch	0.743*	0.775	1.040**	
	(0.446)	(0.896)	(0.472)	
imm_r				4.073*
				(2.392)
<i>N</i>	2075	2075	2040	2040
CZ FE	No	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
First-stage F	.	16.15	.	8.23

Clustered SEs at CZ level. *** p<0.01 ** p<0.05 * p<0.1

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5 What I Learned About the Data

The data required very careful handling. First, the geography crosswalks differ substantially across decades: 1970 uses county group identifiers (CNTYGP97), but 1980 uses a different county group scheme (CNTYGP98), and 1990–2010 use PUMAs. Each crosswalk carries an allocation factor to handle geographic units that straddle multiple commuting zones. Second, the wage variable required substantial cleaning: top-coded incomes and zero or missing hours and weeks account for a large share of observations, particularly in 1970. I also had problems finding 2000 data to be 0 observations, then I had to re-download it separately and merge it, then re-do everything. It made me realize how important it is to make sure all the data are ready in the first place. Besides, the `joinby` merge increases observation counts compared

to the raw extract, because some geographic units overlap multiple commuting zones; using `rawsum` rather than `sum` in the subsequent collapse is essential to recover correct population totals.

6 AI Use Disclosure

I used Grammarly pro to correct my grammar mistakes and adjust some of the expressions. I used Claude (Anthropic) for assistance with Stata code structure, debugging geography merge issues, and learn how to use LaTeX formatting. All empirical decisions, variable construction choices, and interpretations are my own. A log of AI interactions is available upon request.

References

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